

Precalc Lesson 1.2 Homework

Answer Key

1.
 - a. $[0, 10]$
 - b. $[0, 5]$
2.
 - a. All real numbers
 - b. $[-1, 1]$
3.
 - a. $\mathbb{R}, x \neq 3$
 - b. $\mathbb{R}, y \neq 0$
4.
 - a. $-9 \leq x \leq 9$
 - b. $0 \leq y \leq 9$
5.
 - a. $\mathbb{R}$
 - b. $\mathbb{R}$
6. Answers will vary. The graph must have a domain of $[0, 24]$. The range must be within $[0, 100]$. The graph must pass the vertical line test.
7. $\mathbb{R}, x \neq 3, -7$
8.
 - a. Answers will vary; a correct response will give an interval on which it is reasonable to have consumed the hot chocolate. Sample solution: $[0, 45]$
 - b. Answers will vary; a correct response will have 170 as the upper bound and an explanation that the temperature will not decrease past a certain temperature, depending on the temperature of the room. Sample solution: $(70, 170]$.
9.
 - a. $\{p \in \mathbb{N} | 74 \leq p \leq 92\}$
 - b. $1976 \leq C(p) \leq 2408$
10. D